

# EE3621: Digital Signal Processing

## Lecture 17: Short-Time Fourier Transform (STFT) & Spectrogram (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

### 1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins): Motivation: Non-stationary signals.** Why DFT/DTFT fail for time-varying signals.
  - **05:00 – 12:00 (7 mins): STFT Definition & Physical Intuition.** Math formulation and local spectrum interpretation.
  - **12:00 – 17:00 (5 mins): Window Function Role.** Rectangular vs. Hanning/Hamming windows, spectral leakage.
  - **17:00 – 25:00 (8 mins): Time-Frequency Resolution Trade-off.** Heisenberg uncertainty principle.
  - **25:00 – 30:00 (5 mins): Spectrogram.** Definition, parameters, and speech formants example.
  - **30:00 – 34:00 (4 mins): Overlap-Add (OLA) Method.** Perfect reconstruction condition and proof.
  - **34:00 – 37:00 (3 mins): Advanced Views.** Wigner-Ville Distribution and DFT-bank interpretation.
  - **37:00 – 40:00 (3 mins): Applications & Checkpoints.** Radar, ECG, Audio compression. 3 Checkpoint questions.
- 

### 2 Motivation: Non-stationary Signals

The Discrete-Time Fourier Transform (DTFT) and the Discrete Fourier Transform (DFT) are powerful tools. The DTFT is defined as:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad (1)$$

**The Stationarity Assumption:** The underlying assumption of the DFT and DTFT is that the signal  $x[n]$  is **stationary**. This means its statistical properties and frequency content do not change over time. When we compute the standard Fourier transform, we sum over all time. The resulting spectrum  $X(e^{j\omega})$  tells us *what* frequencies exist, but it destroys any information about *when* those frequencies occurred.

**Real-world Non-stationary Signals:**

- **Speech:** Your vocal tract changes shape. A vowel sound has different harmonics than a consonant.
- **EEG (Brainwaves):** Brain electrical activity changes depending on the state of the subject.
- **Audio:** A piano playing a melody produces different notes at different times.

### 3 STFT Definition

To analyze a signal at a specific time  $n$ , we multiply the signal by a window function  $w[m - n]$  centered at time  $n$ . This window isolates a short segment of the signal around time  $n$ . We then take the Fourier transform of this isolated segment.

The Short-Time Fourier Transform (STFT) is defined as:

$$X[n, k] = \sum_{m=-\infty}^{\infty} x[m]w[m - n]e^{-j\frac{2\pi}{N}km} \quad (2)$$

#### Step-by-Step Breakdown:

1.  $x[m]$  is our infinite-length signal.
2.  $w[m - n]$  is the **analysis window** of length  $M$ , shifted to time  $n$ .
3.  $x[m]w[m - n]$  is the windowed signal.
4. The summation is the  $N$ -point DFT of the windowed signal.
5. The result  $X[n, k]$  represents the **local spectrum** of the signal at time  $n$ .

### 4 The Role of the Window Function

The choice of the window function  $w[m]$  is crucial.

#### The Rectangular Window:

$$w_{rect}[m] = \begin{cases} 1, & 0 \leq m \leq M - 1 \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

Because the rectangular window has sharp transitions, its frequency response has a narrow mainlobe but very high sidelobes. This causes **spectral leakage** (Gibbs phenomenon), where energy from one frequency obscures nearby frequencies.

**Hamming Window:** To reduce spectral leakage, we taper the window:

$$w_{hamming}[m] = 0.54 - 0.46 \cos\left(\frac{2\pi m}{M - 1}\right) \quad \text{for } 0 \leq m \leq M - 1 \quad (4)$$

Tapering reduces sidelobes (less leakage) but widens the mainlobe (decreased frequency resolution).

## 5 Time-Frequency Resolution Trade-off

The **Heisenberg Uncertainty Principle** states that for continuous signals:

$$\Delta t \cdot \Delta f \geq \frac{1}{4\pi} \quad (5)$$

For discrete signals, a similar bound limits the bandwidth-duration product:

- **Wide Window (Large  $M$ ):** Excellent frequency resolution, poor time resolution.
- **Narrow Window (Small  $M$ ):** Excellent time resolution, poor frequency resolution.

You cannot simultaneously achieve arbitrarily high resolution in both time and frequency.

## 6 Spectrogram

The **Spectrogram** visualizes the STFT as a 2D magnitude squared plot:

$$\text{Spectrogram}(n, k) = |X[n, k]|^2 \quad (6)$$

Practical parameters include window length (e.g., 20-30 ms for speech), hop size (overlap), and FFT size. In a speech spectrogram, vowels appear as dark bands of energy called **formants**.

## 7 Overlap-Add (OLA) Method & Perfect Reconstruction

To convert an STFT back to the time domain, we use the Overlap-Add method:

$$x[n] = \frac{\sum_{r=-\infty}^{\infty} y_r[n]}{\sum_{r=-\infty}^{\infty} w[n - rH]} \quad (7)$$

**Theorem:** Perfect reconstruction requires  $\sum_r w[n - rH] = C$  (a constant).

**Proof:**

1. Let  $y_r[n] = x[n]w[n - rH]$  be the inverse FFT of frame  $r$ .
2. Summing these overlapping frames:  $\sum_r y_r[n] = \sum_r x[n]w[n - rH] = x[n] \sum_r w[n - rH]$ .
3. If  $\sum_r w[n - rH] = C$ , then  $\sum_r y_r[n] = Cx[n] \implies x[n] = \frac{1}{C} \sum_r y_r[n]$ .

## 8 Alternative Views & Applications

**Wigner-Ville Distribution:** Optimal resolution but introduces artificial cross-terms for multi-component signals.

**DFT-Bank Interpretation:** STFT can be viewed as passing the signal through a uniform bank of modulated bandpass filters.

**Applications:** Speech processing (formants), MP3 audio compression (using MDCT), radar Doppler processing, and ECG analysis.

## 9 Key Formulas

| Concept               | Formula                                               |
|-----------------------|-------------------------------------------------------|
| STFT                  | $X[n, k] = \sum_m x[m]w[m - n]e^{-j\frac{2\pi}{N}km}$ |
| Spectrogram           | $S(n, k) =  X[n, k] ^2$                               |
| Uncertainty Principle | $\Delta t \cdot \Delta f \geq \frac{1}{4\pi}$         |
| OLA Reconstruction    | $x[n] = \frac{\sum_r y_r[n]}{\sum_r w[n - rH]}$       |

## 10 Checkpoint Questions

- Q1:** If you increase the window length  $M$ , what happens to time and frequency resolution?  
**Answer:** Increasing  $M$  spans a longer duration, worsening time resolution but improving frequency resolution (narrower mainlobe).
- Q2:** Why prefer a Hamming window over rectangular?  
**Answer:** Rectangular windows cause severe spectral leakage due to sharp edges. Hamming windows taper smoothly, reducing sidelobes and leakage.
- Q3:** What is the maximum hop size  $H$  for a rectangular window  $w[n] = 1$  ( $0 \leq n \leq 99$ ) to satisfy COLA without division?  
**Answer:**  $H = 100$ . The frames must tile perfectly without gaps or overlapping accumulations to maintain a constant sum of 1.